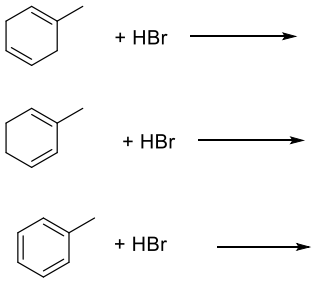
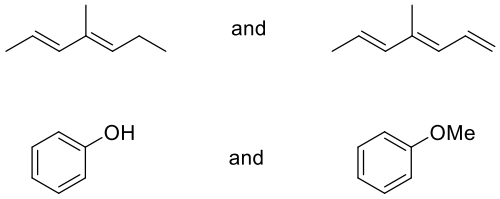
Organic Cumulative Examination
January 23, 2026

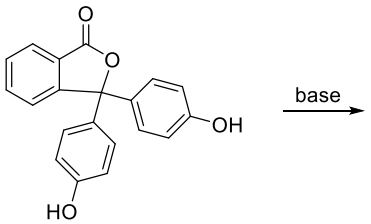
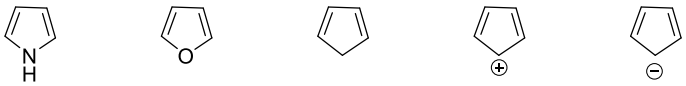
1) Predict the reaction product and complete the following reactions; one equivalent of each reagent is used (30 points)



2) How could you use UV spectroscopy to distinguish between the given compounds in each of the following pairs? (20 points)



3) Phenolphthalein (structure below) is a compound that can change color change depending on the pH of the solution. Solutions of pH values smaller than 8.5 are colorless, and solutions with a pH larger than 8.5 are deep red-purple-pink. Explain the color change (15 points)



4) State the Hückel rules and then indicate for each compound, if it is aromatic, anti-aromatic or not aromatic at all. Give a rationale for each! (35 points total)